

Stellar PSFs Fake Hosts in Little Red Dots

Broad-band point-spread functions built from field stars misassign light when source spectra differ.

arXiv:2608.30459 — Source-conditioned polychromatic PSFs for compact high-redshift nuclei

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Background

Compact high-redshift sources overlap physically distinct components within one diffraction-limited image. Host properties are usually inferred by fitting field-star point-spread functions to the target.

This method assumes the target's spectrum matches the field stars, which is rarely true for little red dots.

What this paper is selling

Using little red dots as a test case, the authors derive the physical and mathematical origin of this mismatch. They demonstrate a source-conditioned polychromatic PSF correction applicable to other compact distant sources.

Key results to remember

- A pure point source can masquerade as an extended host in broad filters.
- The spurious F444W host allocation survives every tested spectral, reference and detector variation.
- Inferred sizes, luminosities and masses of little red dots are systematically biased high.

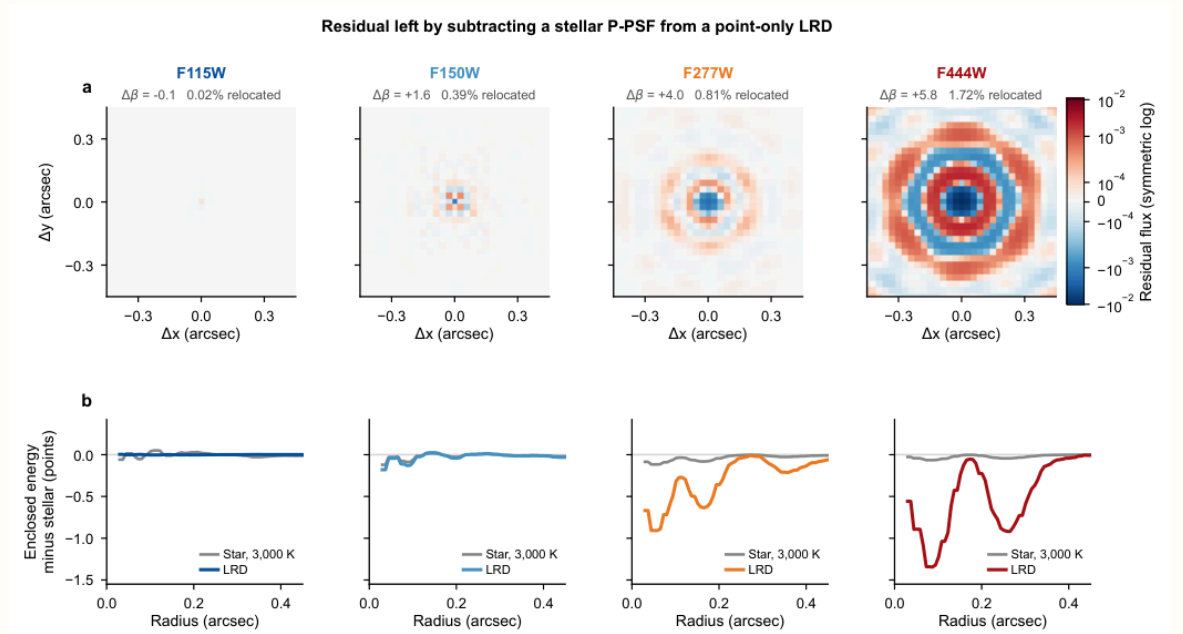


Figure 1 | Subtracting a stellar PSF from a point-only little red dot leaves wavelength-dependent residuals mimicking a Sérsic host.

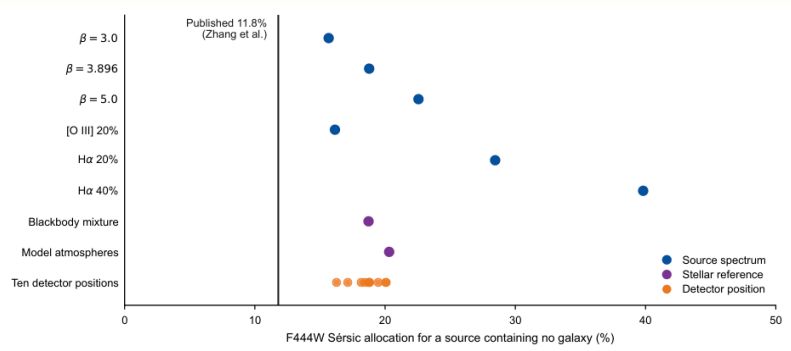


Figure 2 | Every tested nuclear spectrum, stellar reference and detector position yields F444W host fractions exceeding the published 11.8 percent.